

AI POWERED E-COMMERCE CHATBOT

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Abstract - The rapid growth of e-commerce platforms has increased the demand for intelligent and efficient customer interaction systems. This paper presents an AI-powered e-commerce chatbot that integrates semantic search and large language models (LLMs) to enhance customer experience. The system utilizes Sentence Transformers to generate semantic embeddings of product descriptions, which are stored in a PostgreSQL database using the pg vector extension. User queries are processed through a conversational interface and converted into embeddings for similarity matching. The chatbot retrieves relevant products using cosine similarity and generates human-like responses using an open-source LLM. This approach improves recommendation accuracy, personalization, and automated customer support compared to traditional keyword-based systems.

Keywords -AI Chatbot, Semantic Search, Sentence Transformers, pgVector, Large Language Models, E-commerce.

I. INTRODUCTION

With the rapid expansion of online shopping platforms, customer expectations for fast and accurate responses have significantly increased, highlighting the limitations of traditional systems that rely on keyword matching and fail to capture user intent and context. Additionally, rule-based chatbots offer only predefined responses, which reduces user satisfaction and engagement. To overcome these challenges, this paper proposes an AI-powered chatbot that integrates semantic search with conversational AI, enabling a more intelligent and responsive interaction system. The proposed solution utilizes a chat interface connected to a backend that understands user intent through vector embeddings and generates context-aware responses using modern natural language processing techniques and open-source large language models. This approach enhances both product discovery and automated customer support, leading to a more efficient and personalized user experience.

II. CHALLENGES AND OBJECTIVES

User Intent & Response Generation: Understanding ambiguous user queries and generating accurate, context-aware responses using AI models.

Semantic Search Efficiency: Ensuring fast and relevant product retrieval from large datasets using vector-based search methods.

Scalability & Performance: Handling high volumes of users and data efficiently without affecting system performance.

Data Security: Protecting user data and maintaining secure communication within the system.

System Development: Designing an AI-powered chatbot integrated with semantic search and embedding techniques like Sentence Transformers.

Recommendations & Automation: Providing personalized product suggestions and automating customer support using open-source LLMs.

User Experience: Enhancing overall interaction through fast, accurate, and intelligent conversational responses.

III PROBLEM STATEMENT

Many existing e-commerce systems and chatbots face serious limitations when handling user queries and providing accurate responses. In many cases, traditional search systems rely on keyword-based matching, which may seem effective but often fails to understand the actual intent of the user. This can lead to irrelevant search results and poor user experience. Similarly, rule-based chatbots provide fixed responses, which may appear helpful but lack flexibility and context awareness. These limitations make it difficult for users to find the right products quickly and efficiently. In addition, such systems are not capable of handling complex or conversational queries, which further reduces their effectiveness. Another major issue is that traditional systems do not provide personalized recommendations, resulting in generic outputs that do not match user preferences. Moreover, the absence of advanced natural language processing techniques limits the system's ability to learn and improve over time. These issues highlight the need for a more intelligent and efficient system that can understand user intent and deliver accurate, context-aware

responses.

To overcome these problems, the proposed system introduces an AI-powered chatbot that combines semantic search and conversational AI techniques. It uses a Sentence Transformer model to convert both product descriptions and user queries into vector embeddings, enabling the system to understand the semantic meaning of the input rather than relying on simple keywords. The system also applies a vector similarity search mechanism using pgvector to retrieve the most relevant products from the database efficiently. In addition, a Large Language Model (LLM) is used to generate natural and human-like responses, improving user interaction and engagement. The system includes a recommendation module that provides personalized product suggestions based on user queries and behavior. Furthermore, the chatbot integrates a user-friendly chat interface that allows seamless communication between the user and the system. By combining these techniques, the proposed system ensures accurate search results, improved personalization, and efficient automated customer support, thereby enhancing the overall user experience.

Algorithm / Technique	Application Area	Strengths	Weaknesses	References
Sentence Transformer (all-MiniLM-L6-v2)	Embedding generation	Semantic understanding, fast, lightweight	Not domain-specific, limited deep context	Reimers & Gurevych (2019), Sentence-BERT
PGVector	Semantic search in PostgreSQL	Integrated with relational DB, efficient storage	Slower than FAISS for large datasets	PostgreSQL pgvector Documentation
Cosine Similarity	Similarity computation	Simple, effective for vectors	Sensitive to normalization	Salton & McGill (1983), Vector Space Model
Open-source LLM (FLAN-T5 / Mistral)	Response generation	Human-like output, improves UX	Resource intensive, may be generic	Google FLAN-T5 (2022), Mistral AI (2023)
Django Backend	API & backend logic	Scalable, easy integration	Slight overhead	Django Documentation
Fresh Chat	User interface	Real-time, user-friendly	Third-party dependency	Freshworks API Documentation

Table 1: Comparison of algorithms with representation of strength and weakness

IV METHODOLOGY:

A. SYSTEM ARCHITECTURE

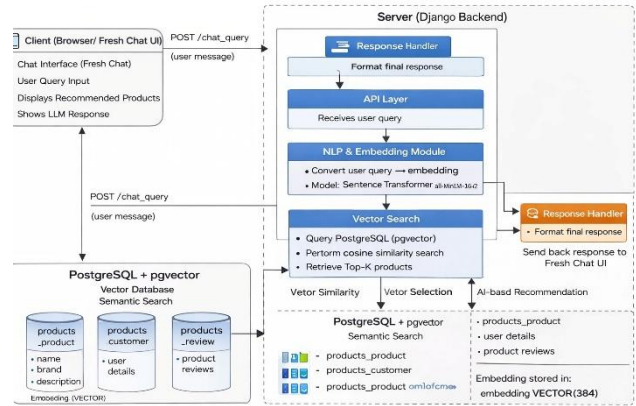


Figure 1: System Architecture

Figure 1 illustrates the architecture of the proposed AI-powered e-commerce chatbot system. The process begins when a user enters a query through the Fresh Chat interface, which is sent to the Django backend. The query is processed in the NLP module and converted into an embedding using a Sentence Transformer model. This embedding is compared with stored product embeddings in PostgreSQL using pgvector through cosine similarity search to retrieve the most relevant products. The results are further refined and passed to an LLM to generate a human-like response. Finally, the formatted response is sent back to the user interface, displaying recommended products and conversational output.

B. DATASET FORMULATION:

The proposed system requires a well-structured dataset to efficiently store and manage product information for accurate retrieval and recommendation. The dataset consists of details such as product ID, product name, category, description, price, ratings, and images. Each product entry is uniquely identified to avoid duplication and ensure consistency within the system. The data is organized in a structured format, allowing easy access and efficient processing during search and recommendation operations.

To enable semantic search, product descriptions are converted into vector embeddings using Sentence Transformers. These embeddings capture the contextual meaning of the text and are stored in a PostgreSQL database using the pgvector extension. This allows the system to perform similarity searches between user queries and stored product data. When a user submits a query, it is also converted into an embedding and compared with stored vectors to retrieve the most relevant products. The system also maintains additional data such as user queries and interaction history, which helps in improving recommendation accuracy over time. All data is stored in structured tables to ensure consistency, scalability, and efficient retrieval. Regular updates can be performed to add new products or modify existing ones. By organizing and maintaining the dataset in this manner, the system ensures efficient data handling, accurate search results, and improved user experience in the e-commerce chatbot.

C. FLOW DIAGRAM:

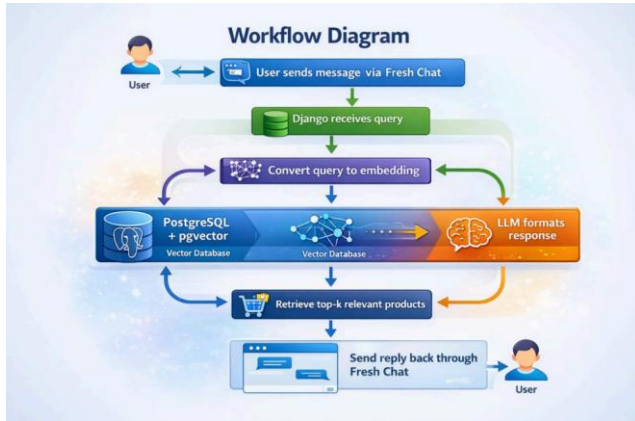


Figure 2: Flow Diagram

Diagram depicting the process flow of the proposed AI-powered e-commerce chatbot system, illustrating how user queries are processed using semantic search and conversational AI to generate intelligent product recommendations.

D. DATA PROCESSING AND EMBEDDING GENERATION

The data processing phase involves preparing and structuring the product dataset for efficient semantic search. Product information such as name, brand, category, description, price, and rating is organized in a structured format. Customer reviews are also integrated with product descriptions to enhance contextual understanding.

The collected data is preprocessed by removing inconsistencies, handling missing values, and normalizing textual content. To capture semantic meaning, the processed text (product description + reviews) is converted into numerical vector representations using a pre-trained Sentence Transformer model (all-MiniLM-L6-v2). These embeddings are 384-dimensional vectors that represent the meaning of the product rather than just keywords.

The generated embeddings are stored in a PostgreSQL database using the pgvector extension. This enables efficient storage and retrieval of high-dimensional vectors, forming the foundation for semantic similarity search.

E. SEMANTIC SEARCH AND RESPONSE GENERATION

The semantic search process begins when a user sends a query through the Fresh Chat interface. The Django backend receives the query and converts it into an embedding using the same Sentence Transformer model to ensure consistency.

The system then performs cosine similarity comparison between the query embedding and the stored product embeddings in the PostgreSQL (pgvector) database. Based on similarity scores, the system retrieves the top-K most relevant products that match the user's intent.

These results are further refined by considering additional factors such as product ratings and relevance to improve recommendation quality. The selected products are then passed to an open-source Large Language Model (LLM), which generates a natural, human-like response.

Finally, the response is formatted and sent back to the user through the Fresh Chat interface, displaying relevant product recommendations along with conversational output. This approach enables context-aware search, improves recommendation accuracy, and enhances user experience by delivering meaningful and personalized results.

V. ALGORITHM:

Sentence Transformer Embedding Algorithm: This algorithm is used to convert product descriptions and user queries into dense vector representations for semantic understanding. The algorithm processes textual data and generates embeddings that capture contextual meaning rather than relying on simple keywords. Each input text is transformed into a high-dimensional vector, enabling accurate comparison between user queries and stored product data. This approach improves the system's ability to understand user intent and retrieve relevant results efficiently.

Vector Similarity Search Algorithm: This algorithm plays a vital role in retrieving relevant products by comparing query embeddings with stored product embeddings. It uses cosine similarity to measure the closeness between vectors and ranks products based on similarity scores. This ensures that the system returns contextually relevant results even when the query does not exactly match product keywords.

LLM-Based Response Generation Algorithm: This algorithm is used to generate natural, human-like responses based on user queries and retrieved product information. It processes the input query along with search results and produces coherent and context-aware replies. The model ensures dynamic interaction by generating responses that are not predefined, improving user engagement and conversational quality.

Recommendation Algorithm: This algorithm is used to provide personalized product suggestions based on user queries and interaction patterns. It analyzes similarities between products, user preferences, and frequently accessed items to recommend relevant products. The system ranks and filters results to display the most useful recommendations, enhancing the overall user experience.

Chat Interface Processing Algorithm: This algorithm handles user interaction by managing message flow between handling of multiple user requests.

VI.IMPLEMENTATION AND RESULTS

The proposed system is implemented as a web-based application following a client–server architecture. The frontend provides an interactive and user-friendly chat interface that allows users to communicate with the chatbot and explore products easily. The backend is developed using Django, which handles core functionalities such as query processing, embedding generation, similarity search, and response generation. It also manages communication between different modules efficiently.

The implementation begins with storing product data in a PostgreSQL database integrated with the pgvector extension. Product descriptions are converted into vector embeddings using a Sentence Transformer model and stored for efficient retrieval. When a user submits a query through the chat interface, it is converted into an embedding and compared with stored vectors using cosine similarity. The system retrieves the most relevant products based on similarity scores.

After retrieving the results, a Large Language Model (LLM) generates a natural and context-aware response by combining the user query and product information. The chatbot then displays the generated response along with recommended products in real time. The system also maintains user interaction history to improve recommendation accuracy and personalization over time.

The results demonstrate that the proposed system provides more accurate and relevant product recommendations compared to traditional keyword-based search systems. It improves user engagement by delivering conversational and context-aware responses. Additionally, the system reduces the need for manual customer support by automating query handling. Overall, the implementation proves to be efficient, scalable, and effective for real-time e-commerce applications, enhancing both search accuracy and user experience.

The output of the applications as shown:

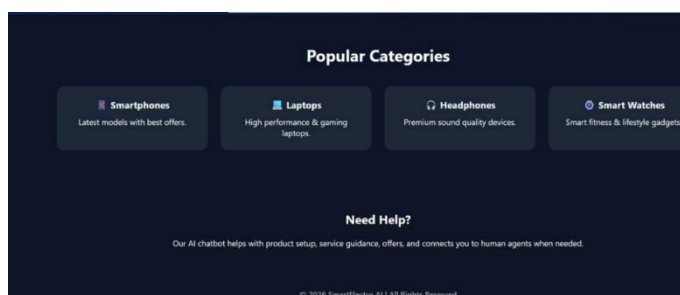
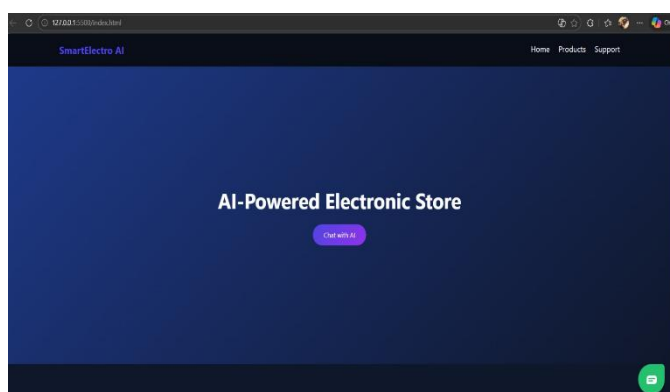


Figure 1: Homepage of AI-Powered E-Commerce System

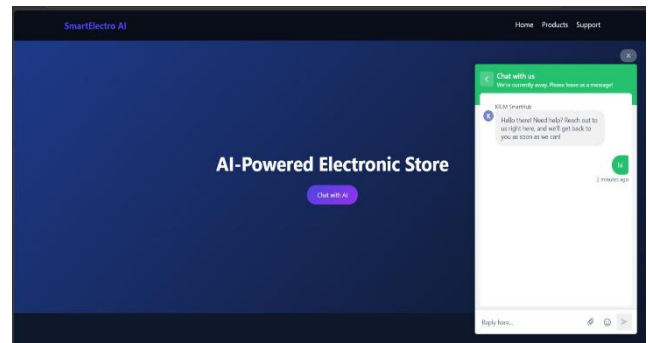


Figure 2: Chatbot Interface Integration

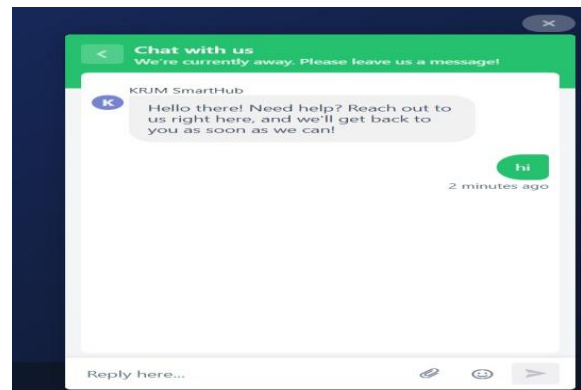


Figure 3: User Interaction with Chatbot

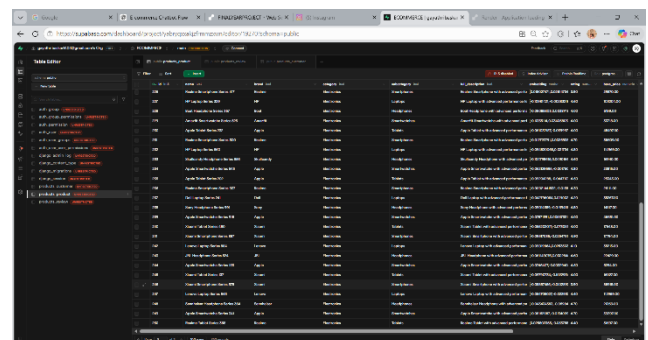


Figure 4: Database Table in PostgreSQL

VII. CONCLUSION

In this conclusion, an AI-powered e-commerce chatbot has been developed to improve product search and customer interaction. The system focuses on understanding user queries more effectively by using semantic search and natural language processing techniques. Users can interact with the chatbot through a simple chat interface, where queries are processed and relevant products are retrieved efficiently. The integration of Sentence Transformers enables accurate understanding of user intent, while the use of vector similarity search ensures precise and relevant results.

The application further enhances user experience by generating natural and context-aware responses using a Large Language Model (LLM). It also provides personalized product recommendations, helping users discover suitable products quickly. Additionally, the system maintains structured data and interaction history, which supports better performance and continuous improvement over time.

Overall, the system ensures efficient data handling, accurate recommendations, and automated customer support. It provides a scalable, reliable, and user-friendly solution for modern e-commerce platforms. This approach improves search accuracy, enhances user engagement, and reduces manual effort, making it suitable for real-world applications in intelligent online shopping systems.

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